

Introduction to Data Analysis

Midterm, Nov 1, 2023 - Time allowed: 2 hours

Course Questions:

Q1) Which of the following measures is least affected by outliers?

- a) Mean
- b) Median
- c) Mode

Q2) In the case of a left-skewed distribution:

- a) Mean > Median > Mode
- b) Mode > Median > Mean
- d) Mode > Mean > Median

Q3) Which of the following scenarios might be represented by a binomial distribution?

- a) Car colors in a parking place
- b) The heights of sunflowers in a field
- c) The number of heads when flipping a coin three times

Q4) Which of the following is a key assumption of the Central Limit Theorem?

- a) The population distribution is not normal
- b) The samples are dependent
- c) The samples are random and independent

[3 marks]

Exercise 1

If you flip a fair coin 15 times and roll a fair six-sided die 4 times,

- What is the probability of getting exactly 11 heads and rolling two 6's?
- Calculate the probability of obtaining at least 10 heads or at least one 6.
- Given that at least one 6 was rolled, what is the probability of obtaining exactly 11 heads?

[2 marks]

Exercise 2

Suppose a survey of 1000 drivers in a metropolitan area during a 3-year period was taken. The following results were found.

Age Group	18-25	26-39	40-55	55+
0-1 accidents	100	150	250	75
2-3 accidents	150	25	125	25
3+ accidents	50	25	25	0

We randomly select a driver from the group. What are the probabilities of the following events:

- the driver is in the 40-55 age group, given they have more than 3 accidents.
- the driver had 2-3 accidents, given they are in the 26-39 age group.
- the driver had 0-1 accidents, given they are in the 40-55 age group.
- the driver is in the 40-55 age group, given they had 0-1 accidents.

[2 marks]

Exercise 3

A retail store analyzes the age of customers visiting the store to improve its marketing strategy. The ages recorded are as follows:

23,25,22,30,35,32,40,42,38,37,45,50,55,60,65,32,38,40,35,30,22,28,35,37,40,50,55,42

- Calculate the mean, median, mode, range and interquartile range of the ages.
- Create a box-and-whisker plot and identify if there are any outliers in age. If yes, how might these outliers impact the marketing strategy?

[2 marks]

Exercise 4

An airport screens bags for forbidden items, and an alarm is supposed to be triggered when a forbidden item is detected. Suppose that 7% of bags contain forbidden items. If a bag contains a forbidden item, there is a 94% chance that it triggers the alarm. If a bag doesn't contain a forbidden item, there is an 8% chance that it triggers the alarm. Given a randomly chosen bag triggers the alarm, what is the probability that it contains a forbidden item?

[2 marks]

Exercise 5

We would like to design a fair game. The game consists of rolling two dice; if the two numbers that turn up differ, you lose one dollar; if the two numbers are the same, you win N dollars.

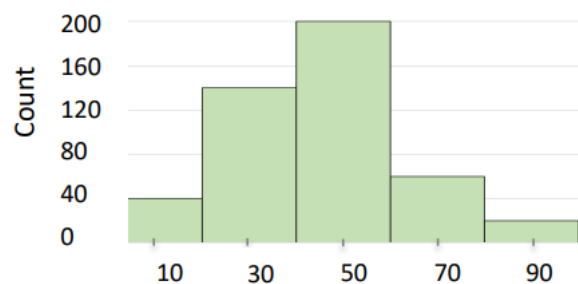
- Find the value of N which will make the game fair
- Using the results of a), calculate the standard deviation of the winning/loss?

[2 marks]

Exercise 6

Consider the histogram below, which corresponds to data collected about a continuous-valued random variable, X.

- Determine the size of the data sample.
- Estimate the median of X.
- Estimate Q1, Q3 and IQR of X.
- Sketch the box-plot of X.
- Estimate the mean of X.



[3 marks]

Study vs Relaxation Time Analysis with R

You've been tracking your study and relaxation hours over a week to better understand your time management. Here's your data:

Study Hours	Relaxation Hours
- Monday you studied for 5 hours	- Monday you relaxed for 2 hours
- Tuesday you studied for 6 hours	- Tuesday you relaxed for 1 hour
- Wednesday you studied for 3 hours	- Wednesday you relaxed for 4 hours
- Thursday you studied for 7 hours	- Thursday you relaxed for 1 hour
- Friday you studied for 4 hours	- Friday you relaxed for 3 hours

Q1. Setting Up the Vectors:

- Assign the hours for 'Study Hours' and 'Relaxation Hours' to the variables ``study_vector`` and ``relax_vector`` respectively.
- Using the days of the week, create a variable ``days_vector``. Use this variable to set the names for both ``study_vector`` and ``relax_vector``.

Q2. Total Hours Analysis:

- Assign to the variable ``total_daily_hours`` the total hours (study + relaxation) for each day.
- Calculate your total study hours for the week and assign it to ``total_study``.
- Similarly, compute the total relaxation hours for the week and assign it to ``total_relax``.
- Compute the ``total_week_hours`` by combining the totals for both study and relaxation.

Q3. Detailed Study and Relaxation Analysis:

- Check if your total study hours for the week are greater than your relaxation hours. Assign the result to ``more_study_than_relax``. What does this indicate about your week?
- Determine on which days your study hours exceeded 5 hours and store the boolean vector in ``intensive_study_days``.
- Extract the hours from the days where relaxation exceeded 2 hours and store them in ``long_relax_days``.

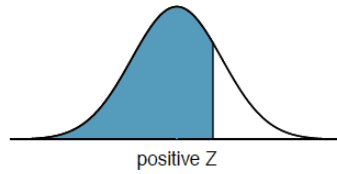
Q4. Specific Days and Averages:

- Assign the study hours of Wednesday to the variable ``study_wednesday``.
- Extract the study hours of Tuesday through Thursday and assign to ``study_midweek``.
- Extract the relaxation hours from Tuesday to Friday and assign to ``relax_endweek``.
- Compute the average study hours for the first three weekdays and store in ``average_initial_study``.

[4 marks]

APPENDIX: PROBABILITY TABLES

NORMAL PROBABILITY TABLE



Z	Second decimal place of Z									
	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

*For $Z \geq 3.50$, the probability is greater than or equal to 0.9998.

Useful formulas

Sample mean: $\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$

Variance: $s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$

Standard deviation: $s = \sqrt{s^2}$

Interquartile range (IQR): $IQR = Q_3 - Q_1$

Whiskers: max upper whisker reach = $Q_3 + 1.5 \times IQR$
max lower whisker reach = $Q_1 - 1.5 \times IQR$

General addition rule: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

Checking for independence: $P(A \text{ occurs, given that } B \text{ is true}) = P(A | B) = P(A)$
 $P(A|B) = \frac{P(A \text{ and } B)}{P(B)} = \frac{P(A) \times P(B)}{P(B)} = P(A)$

Independent events: $P(\text{at least one}) = 1 - P(\text{none})$
 $P(A \text{ or } B) = P(A) + P(B)$
 $P(A \text{ and } B) = P(A) \times P(B)$

Conditional probability: $P(A|B) = \frac{P(A \text{ and } B)}{P(B)}$

General multiplication rule: $P(A \text{ and } B) = P(A|B) \times P(B)$

Bayes formula: $P(A | B) = \frac{P(B | A) \cdot P(A)}{P(B)}$

Expected value(mean): $\mu = E(X) = \sum_{i=1}^k x_i P(X = x_i)$

Variability: $\sigma^2 = \text{Var}(X) = \sum_{i=1}^k (x_i - E(X))^2 P(X = x_i)$

Standard Deviation: $\sigma = SD(X) = \sqrt{\text{Var}(X)}$

Linear combination of variables: $aX + bY$

Average values: $E(aX + bY) = a \times E(X) + b \times E(Y)$

Variability: $V(aX + bY) = a^2 \times V(X) + b^2 \times V(Y)$

More rules: $E(X + X) = E(X) + E(X) = 2E(X)$
 $E(2X) = 2E(X)$
 $\text{Var}(2X) = 2^2 \text{Var}(X) = 4 \text{Var}(X)$
 $\text{Var}(X + X) = \text{Var}(X) + \text{Var}(X) = 2 \text{Var}(X)$

Z score: $Z = \frac{\text{observation} - \text{mean}}{SD}$

Geometric distribution:

Probability of success: p Probability of failure: $1-p$ Geometric probability: $P(\text{success on the } n^{\text{th}} \text{ trial}) = (1-p)^{n-1}p$
Mean: $\mu = \frac{1}{p}$ Standard deviation: $\sigma = \sqrt{\frac{1-p}{p^2}}$

Binomial distribution:

Probability of a single scenario: $P(\text{single scenario}) = p^k (1-p)^{(n-k)}$ Choose function (number of scenarios): $\binom{n}{k} = \frac{n!}{k!(n-k)!}$
Binomial probability: $P(k \text{ successes in } n \text{ trials}) = \binom{n}{k} p^k (1-p)^{(n-k)}$ Mean: $\mu = np$ Standard deviation: $\sigma = \sqrt{np(1-p)}$
Negative binomial distribution: $P(k^{\text{th}} \text{ success on the } n^{\text{th}} \text{ trial}) = \binom{n-1}{k-1} p^k (1-p)^{n-k}$

Operation	R operator	Example
Create a vector	<code>vector<- c(elem1, elem2, elem3,...)</code>	numeric vector <- c(1, 10, 49) character vector <- c("a", "b", "c") boolean vector <-c(TRUE,FALSE,TRUE)
Naming a vector	<code>names(vector) <- c("name1", "name2")</code>	some vector <- c("Johnny", "Poker Player") names(some vector) <- c("Name", "Profession") some_vector
Sum of two vectors	<code>vector1+vector2</code>	A_vector <- c(1, 2, 3) B_vector <- c(4, 5, 6) total_vector <- A_vector+B_vector
sum of all elements of a vector	<code>sum(vector)</code>	sum(A_vector)
average of a vector	<code>mean(vector)</code>	mean(A_vector)
Comparing two vectors	<code>vector1 > vector2</code>	A_vector > B_vector
Vector selection	<code>vector[i]</code>	A_vector A_vector[1] A_vector[2]
	<code>vector["name"]</code>	some_vector <- c("Johnny", "Poker Player") names(some_vector) <- c("Name", "Profession") some_vector["Name"]
select multiple elements from a vector	<code>vector[c(indice 1,indice 2,...)]</code>	C_vector <- c(11,33,55,7,9) C_vector[c(1,3)]
	<code>vector[c(indice 1:indice 2)]</code>	C_vector[c(1:3)]
	<code>Vector[c("name1", "name2",...)]</code>	some_vector[c("Name", "Profession")]
Selection by comparison	<code>vector<-compraision</code> <i>The (logical) comparison operators known to R are:</i> <i>< for less than</i> <i>> for greater than</i> <i><= for less than or equal to</i> <i>>= for greater than or equal to</i> <i>== for equal to each other</i> <i>!= not equal to each other</i>	A_vector <- c(1, 2, 3) B_vector <- c(4, 5, 6) C_vector<-A_vector>B_vector D_vector<- A_vector!=B_vector E_vector<-A_vector > 2